

MA.912.AR.2.4

Benchmark

MA.912.AR.2.4 Given a table, equation or written description of a linear function, graph that function, and determine and interpret its key features.

Benchmark Clarifications:

Clarification 1: Key features are limited to domain, range, intercepts and rate of change.

Clarification 2: Instruction includes the use of standard form, slope-intercept form and point-slope form.

Clarification 3: Instruction includes cases where one variable has a coefficient of zero.

Clarification 4: Instruction includes representing the domain and range with inequality notation, interval notation or set-builder notation.

Clarification 5: Within the Algebra I course, notations for domain and range are limited to inequality and set-builder notations.

Connecting Benchmarks/Horizontal Alignment

- MA.912.F.1.3
- MA.912.F.1.5
- MA.912.F.1.6

Terms from the K-12 Glossary

- Coordinate Plane
- Domain
- Function Notation
- Range
- Rate of Change
- Slope
- x -intercept
- y -intercept

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.4

Next Benchmarks

- MA.912.NSO.4.2
- MA.912.AR.9.8, MA.912.AR.9.9, MA.912.AR.9.10

Purpose and Instructional Strategies

In grade 8, students graphed two-variable linear equations given a written description, a table, or an equation in slope-intercept form. In Algebra I, students graph linear functions from equations in other forms, as well as tables and written descriptions, and they determine and interpret the domain, range and other key features. In later courses, students will graph and solve problems involving linear programming, systems of equations in three variables and piecewise functions.

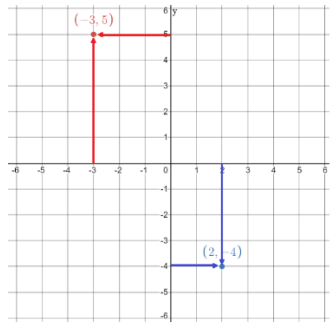
- Instruction includes representing domain and range using words, inequality notation and set-builder notation.
 - Words
If the domain is all real numbers, it can be written as “all real numbers” or “any value of x , such that x is a real number.”
 - Inequality Notation
If the domain is all values of x greater than 2, it can be represented as $x > 2$.
 - Set-Builder Notation
If the domain is all values of x less than or equal to zero, it can be represented as $\{x|x \leq 0\}$ and is read as “all values of x such that x is less than or equal to zero.”
- Within this benchmark, linear two-variable equations include horizontal and vertical lines. Instruction includes writing horizontal and vertical lines in the form $y = 3$ and $x = -4$ and as $0x + 1y = 3$ and $1x + 0y = -4$, respectively. Students should understand that vertical lines are not linear functions, but rather linear two-variable equations.
 - Discussions about this topic are a good opportunity to foreshadow the use of horizontal and vertical lines as common constraints in systems of equations or inequalities.
- Instruction includes the use of x - y notation and function notation.
- Instruction includes the use of appropriately scaled coordinate planes, including the use of breaks in the x - or y -axis when necessary.

Common Misconceptions or Errors

- Students may express initial confusion with the meaning of $f(x)$ for functions written in function notation.

Strategies to Support Tiered Instruction

- Teacher provides equations in both function notation and x - y notation written in slope-intercept form and models graphing both forms using a graphing tool or graphing software (MTR.2.1).
 - For example, $f(x) = \frac{2}{3}x + 6$ and $y = \frac{2}{3}x + 6$, to show that both $f(x)$ and y represent the same outputs of the function.
- Teacher provides instruction using a coordinate plane geoboard to provide students support in graphing a linear function.
 - For example, given the y -intercept and slope of line, teacher first puts a peg at the y -intercept. Then, the teacher models using the slope to graph a second point by moving a second peg from the y -intercept “up and over” (or “down and over”) to another point. Students can check their second point by plugging it into the given equation of the line. If the equation makes a true statement, then the student has graphed the second point correctly. If the equation makes a false statement, then the student has not graphed the second point correctly.
- For students who need extra support in plotting points, teacher provides instruction using a coordinate plane geoboard to find the x -value on the x -axis and the y -value on the y -axis. Then, the teacher models moving to find the coordinate point, where the x - and y -value meet.
 - The graph below shows how a teacher could model plotting the points $(-3, 5)$ and $(2, -4)$.



Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.7.1)

There are a total of 549 seniors graduating this year. The seniors walk across the stage at a rate of 42 seniors every 30 minutes. The ceremony also includes speaking and music that lasts a total of 25 minutes.

Part A. Write a function that models this situation.

Part B. What is the slope of the function you created? How does that translate to this situation?

Part C. Graph the function you created in Part A. What is the feasible domain and range for this situation?

Part D. For how many hours will the graduation ceremony take place?

Instructional Items

Instructional Item 1

Part A. Graph the function $g(x) = -3.6x + 7$.

Part B. Identify the domain, range and x - and y -intercepts of the function.

Instructional Item 2

Graph the linear function represented in the table below.

x	-3	1	7	9	18
$f(x)$	-2.62	-1.66	-0.22	0.26	2.42

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*